

## OUTPUT :

|   | Hours_Studied | Attendance | Assignment_Score | Midterm_Score | Final_Score |
|---|---------------|------------|------------------|---------------|-------------|
| 0 | 1             | 60         | 55               | 50            | 52          |
| 1 | 2             | 65         | 58               | 55            | 57          |
| 2 | 3             | 70         | 60               | 58            | 60          |
| 3 | 4             | 75         | 65               | 62            | 64          |
| 4 | 5             | 80         | 68               | 65            | 68          |

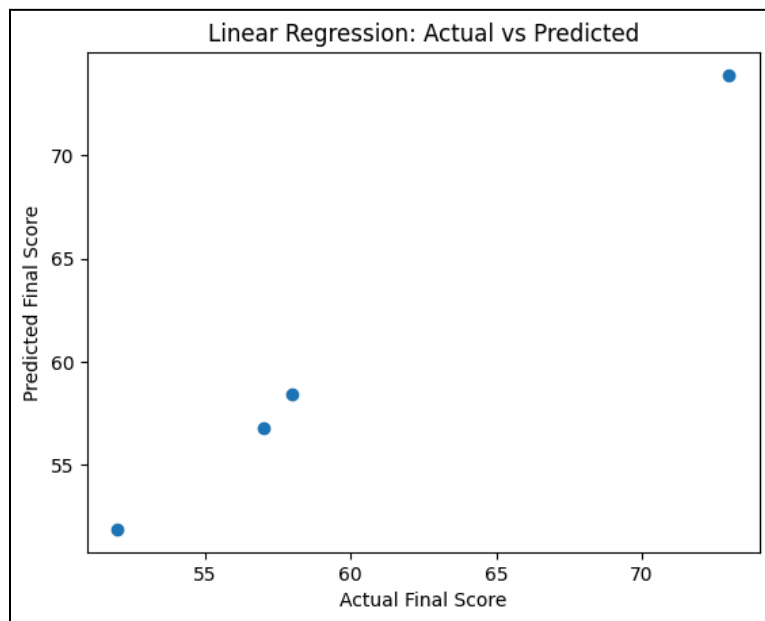
### ===== LINEAR REGRESSION RESULTS =====

MAE: 0.41749050153075906

MSE: 0.2612765457256682

RMSE: 0.5111521747245806

R2 Score: 0.9957516008825095



Pass/Fail Distribution:

Pass

1 13

0 7

Name: count, dtype: int64

### ===== LOGISTIC REGRESSION RESULTS =====

Accuracy: 1.0

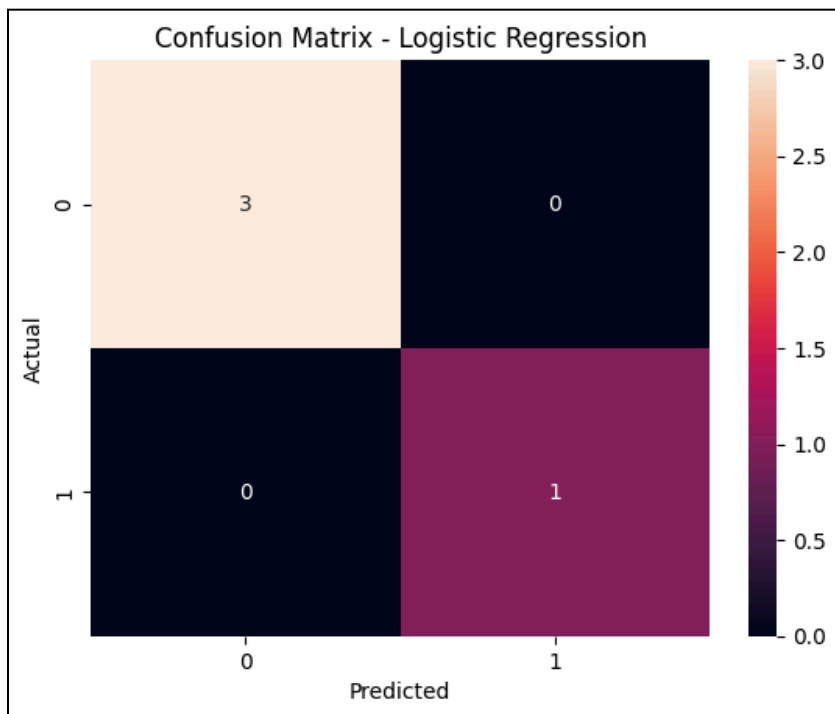
Confusion Matrix:

[[3 0]

[0 1]]

### Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 1.00      | 1.00   | 1.00     | 3       |
| 1            | 1.00      | 1.00   | 1.00     | 1       |
| accuracy     |           |        | 1.00     | 4       |
| macro avg    | 1.00      | 1.00   | 1.00     | 4       |
| weighted avg | 1.00      | 1.00   | 1.00     | 4       |



### ===== HYPERPARAMETER TUNING =====

Best Parameters: {'C': 0.1, 'solver': 'lbfgs'}

Best Cross-Validation Accuracy: 1.0